ELECTRIC POWER STEERING (EPS) MOTOR

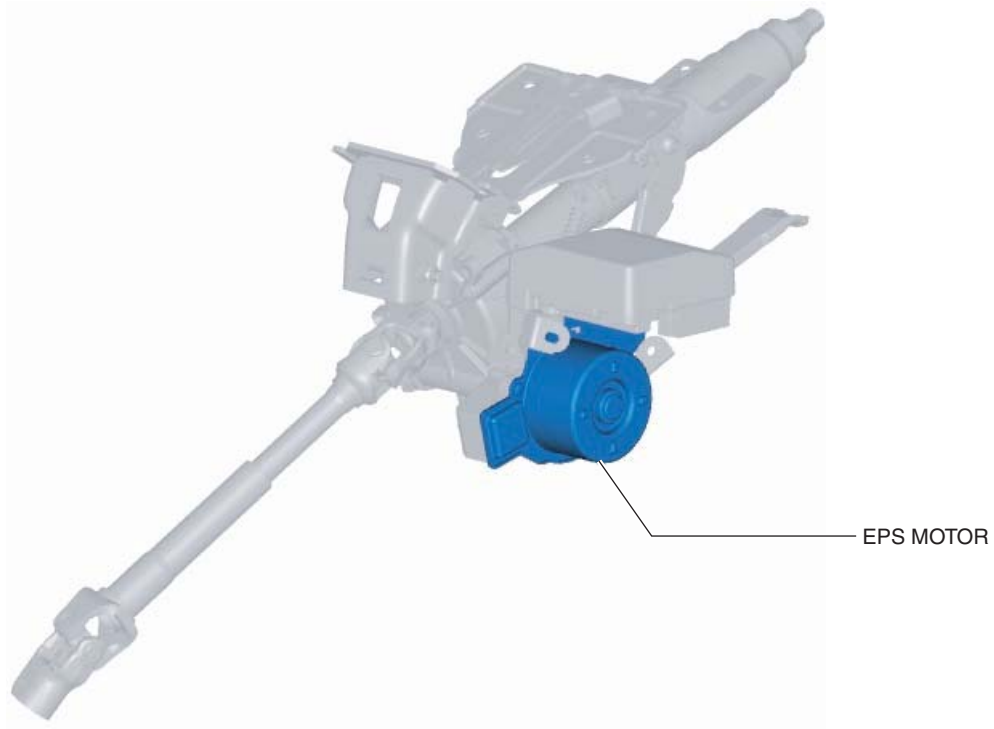
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Purpose, Function

- The EPS motor is driven by the control current from the EPS control module (CM) and generates assist torque to manual steering mechanism.

Construction

- The EPS motor is integrated with the steering column.
- The EPS motor is a brushless AC motor which consists of a three-phase, star-shaped connection stator, a permanent magnet rotor, and a motor rotation angle sensor.



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Operation

- Electrical current from the EPS CM is applied to the stator, and the motor axis integrated with the rotor is rotated.
- The EPS motor has a motor rotation angle sensor which detects the rotor position, and it outputs a rotor position signal to the EPS CM.
- The EPS CM switches the current flowing to each stator based on the rotor position signal from the motor rotation angle senor.

Fail-safe

DTC No.	Fail-safe function
U200D:49	• Control disabled • Power steering malfunction indicator light: Illuminated
U2011:49	
U2011:61	